

[Figure 3 — placeholder]

Square light-teal domain with background shear modulus μ^0 , containing an irregular pink inclusion region with shear modulus μ^1 and boundary $\partial\Omega^1$. Eight control points $\mathbf{P}_1^1, \mathbf{P}_2^1, \dots, \mathbf{P}_8^1$ are distributed around the boundary, connected to each other by dashed blue segments (the control polygon) and smoothly interpolated by an orange closed B-spline curve. Eight radial distances $d_1^1, d_2^1, \dots, d_8^1$ connect the inclusion center (x_0^1, y_0^1) to each control point, illustrating the polar parameterization used in Eq. (7) and Eq. (10).

Fig. 3. The nonhomogeneous shear modulus distribution of an inclusion embedded composite.

[Figure 4 — placeholder]

Graph of $H(z) = 0.5 \times (1 + \tanh(z/\varepsilon))$ (the smoothed Heaviside from Eq. (13)) plotted over $z \in [-0.5, 0.5]$, with $H(z) \in [0, 1]$ on the vertical axis. Four curves show the effect of decreasing ε :

- $\varepsilon = 0.01$ sharpest transition (nearly a true step)
- $\varepsilon = 0.05$ moderate transition
- $\varepsilon = 0.10$ smoother transition
- $\varepsilon = 0.15$ smoothest transition

All four curves pass through $(0, 0.5)$. As $\varepsilon \rightarrow 0$ the approximation converges to the true Heaviside; as ε grows the transition band widens.

Fig. 4. An approximation of Heaviside function.

$$\text{Reg}(\mu) = \frac{1}{2} \alpha \int_{\Omega} \sqrt{|\nabla \mu|^2 + c^2} d\Omega \quad (15)$$

For the layered structure, $\mathbf{r}^i = [r_1^i(t_1), r_2^i(t_2), \dots, r_n^i(t_n)]$ is the geometric parameter vector representing the distance between the i -th interface and C^0 at the location t_k . For the structure with inclusions embedded, the geometric parameter vector $\mathbf{r}^i = [r_1^i, r_2^i, \dots, r_n^i] = [x_0^i, y_0^i, d_1^i, \dots, d_{n-2}^i]$ is consisting of the coordinates of the center and distances between the control points to the center of i -th component. In Eq. (14), the discrepancy between the computed displacement field and measured displacement field on partial surface $\partial\Omega_{i_{\text{meas}}}$ is minimized in L_2 norm. The computed displacement field is obtained by solving a forward problem using the finite element method [32]. Due to the influence of noise in measurements, we introduce a total variation diminishing regularization (TVD) term (the second term in Eq. (14)), and the specific form is given by Eq. (15). α is the regularization factor which controls the significance of the regularization on the objective function. c is a small constant to avoid singularities when evaluating the gradient of the regularization term with respect to optimization variables. It should be noted that there exist alternative forms for the regularization term apart from the domain integral forms. Nevertheless, we have chosen to adhere to the use of domain integrals due to practical considerations. In particular,

the explicit inverse solver has been developed based on the established framework of the existing implicit inverse solver. This continuity in methodology ensures consistency and facilitates seamless integration between the two solvers [33]. The specific value of the regularization factor is chosen based on the smoothness criteria [22].